

Plotting a Series of Points with scatter()

To plot a series of points, we can pass `scatter()` separate lists of *x*- and *y*-values, like this:

```
scatter import matplotlib.pyplot as plt
_squares.py
x_values = [1, 2, 3, 4, 5]
y_values = [1, 4, 9, 16, 25]

plt.style.use('seaborn')
fig, ax = plt.subplots()
ax.scatter(x_values, y_values, s=100)

# Set chart title and label axes.
--snip--
```

The `x_values` list contains the numbers to be squared, and `y_values` contains the square of each number. When these lists are passed to `scatter()`, Matplotlib reads one value from each list as it plots each point. The points to be plotted are (1, 1), (2, 4), (3, 9), (4, 16), and (5, 25); Figure 15-6 shows the result.

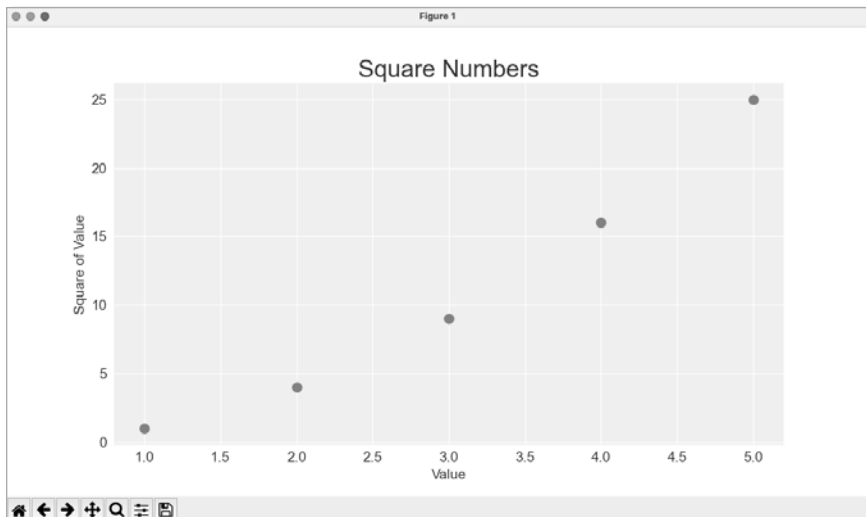


Figure 15-6: A scatter plot with multiple points

Calculating Data Automatically

Writing lists by hand can be inefficient, especially when we have many points. Rather than writing out each value, let's use a loop to do the calculations for us.

Here's how this would look with 1,000 points:

```
scatter import matplotlib.pyplot as plt
_squares.py
❶ x_values = range(1, 1001)
y_values = [x**2 for x in x_values]
```